Luke Eyles

Mechanical and Mechatronics Engineering Student
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Education

Bachelor of Mechanical and Mechatronic Engineering, 2017 – 2022 (Expected)
Diploma in Professional Engineering Practice
University of Technology Sydney

- GPA: 6.55/7.00, High Distinction average.
- Mechanical and Mechatronic Design: Worked in a team in collaboration with Thales Australia on a project researching underwater robot communication.
- Robotics: Led team to program a robot arm to play tic-tac-toe with a person [video link].
- Programming for Mechatronic Systems: Programmed a simulated aircraft in C++ to intercept enemy aircraft by predicting their trajectory.

Experience

Technology Consulting Intern Nov 2020 – Present
Isle Utilities

- Performing comprehensive technology scans on areas including water conservation, leakage management, and waste to energy.
- Creating communication materials for clients including reports, emails, videos, and presentations.
- Drafted trial proposals for collaborative trials of technologies.

Engineering Intern in Research and Development Department Aug 2019 – Feb 2020
Jenkins Engineering Defence Systems

- Created Signal Generator GUI application in MATLAB that interfaced with software defined radios to transmit pre-set or user defined signals.
- Developed Signal Analyser GUI in MATLAB, allowing a quick overview of signal characteristics, or in-depth analysis.
- Created system to record sensor data from an IMU to a single board computer using C++.

Assistant Lab Tutor for Introduction to Electrical Engineering Mar 2018 – Jun 2018
University of Technology Sydney

- Explained electrical concepts and components.
- Debugged students' circuits and walked through how to fix the error.

Skills

Technical Skills

- **C++:** Object oriented programming, using threads to process and send multiple streams of data, compiling using CMake.
- **MATLAB**: Image processing, signal processing, robotics using Peter Corke's Robotics Toolbox and ROS Toolbox, GUI programming using App Designer.
- **Robot Operating System (ROS)**: Controlling real and simulated robots in MATLAB and C++, processing incoming sensor messages, visualising using RViz.
- **Microcontrollers**: Interfacing with sensors, controlling LCD screen and motors, using timers.
- **Git**: Version control and collaboration.
- **CAD***:*
    - **Solidworks**: Designing parts, performing FEA, creating assemblies, creating and interpreting technical drawings.
    - **Blender**: Creating meshes, using modifiers, applying procedural and image-based textures.
- **Python**
- **Microsoft Excel:** Recording data, using formulas, creating charts, sorting and filtering data.
- **Microsoft Office**

Core Skills

- **Team leadership:** Successfully divided work between team members, coordinated different parts of the project, organised team meetings, and ensured deadlines were met for the major robotics project in Sensors and Control.
- **Visual communication skills**: Illustrated diagrams to communicate concepts to others clearly in presentations, reports, and tutoring.
- **Written and verbal communication skills:** Listened to requirements and drafted formal project specification, participated in team meetings by contributing new ideas and adding to others' ideas, wrote clear and concise reports, and gave project updates and presentations when working on the Mechanical and Mechatronic Design project with Thales Australia.
- **Time management:** Tracked tasks for team and individual projects using project management software, set deadlines for project milestones, worked flexibly to changing requirements, and used calendar to manage deadlines for work and university.

References

Available on request.